Ocean Ecosystems

RESEARCH Open Access

Similar sensitivity and resilience to marine heatwaves of giant kelp (*Macrocystis pyrifera*) sporophytes from the northern and southern edges of their distribution in Aotearoa New Zealand

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Abstract

Background Ocean warming and marine heatwaves have contributed to a global decline in kelp canopy cover, threatening the persistence of important kelp forest ecosystems in many locations. Kelp populations at the warm edge of their range can become more resilient to extreme temperatures through local thermal adaptation, but such adaptation has not occurred in some regions. In this study, five populations of the giant kelp *Macrocystis pyrifera* from the northern and southern edges of the species' range in Aotearoa New Zealand were exposed to simulated 22-day marine heatwaves. Juvenile sporophytes were exposed to either 18 °C or 22 °C heatwaves, or a 14 °C control. The experiment also assessed whether the long-term storage of gametophytes under red light, a technique that is commonly used to create germ banks, could affect sporophyte performance.

Results It was found that the 22 °C heatwave treatment caused 64% mortality and a significant reduction in blade growth rates across all kelp cultures, whereas all cultures were comparatively resistant to the 18 °C treatment, with just 23% mortality during the heatwave. Chemical composition of sporophytes was altered in both heatwave treatments, with total carbon content, carbon:nitrogen ratio, and $\delta^{15}N$ values increasing significantly. There was little evidence that long-term storage of kelp gametophytes under red light impacts sporophyte survival or growth. Although survival rates and chemical composition differed between some populations, there was no consistent evidence overall for significant differences in thermal tolerance between northern and southern *M. pyrifera* cultures.

Conclusions Based on these findings, there is little evidence for local thermal adaptation among *M. pyrifera* sporophytes in New Zealand, and recent thermal history and acclimatisation may contribute more towards thermal tolerance than long-term adaptation. Genetic connectivity between kelp populations might also erode local adaptation. If marine heatwave intensities continue to increase as the oceans warm, the survival, growth, and recruitment of juvenile sporophytes are likely to be reduced in *M. pyrifera* populations throughout New Zealand, especially in regions such as Fiordland that are exposed to very high temperature anomalies and cumulative stress. We recommend

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that future studies of thermal tolerance in *M. pyrifera* focus on within-population variability in thermal resilience and assess the potential effectiveness of thermal priming for vulnerable populations.

Keywords Kelp, Macroalgae, Marine heatwaves, Ocean warming, Adaptation, Acclimatisation, Resilience

1 Background

Kelp forests are biogenic habitats that are widespread in temperate coastal regions of every continent except Antarctica [73]. Kelps modify the chemical composition [14] and structural and hydrodynamic characteristics of their environment, thus functioning as ecosystem engineers [6, 49, 115]. Kelp forests provide habitat for a diverse range of flora and fauna, including numerous culturally and commercially important species [25, 52, 115], whilst kelps themselves are culturally significant to many indigenous communities [118, 121]. Additionally, kelp forests provide ecosystem services [38] including carbon and nutrient cycling [11, 44], increasing local pH [14, 28], and reducing coastal erosion [81].

Climate change currently threatens many marine ecosystems, with kelp forests being particularly vulnerable [27, 109, 124] and experiencing a global decline as a result [73]. Elevated water temperatures can cause thinning and damage to connective tissues in kelp blades, leading to rapid erosion [41, 64, 107]. Ocean warming has driven a global increase in the frequency and severity of marine heatwaves (MHWs), extreme events in which ocean temperatures exceed the 90th percentile for 5 days or more [30, 59, 108]. During MHW events, water temperatures can rapidly exceed critical thresholds for sensitive marine organisms, causing widespread mortality [110]. This restricts species' ability to acclimatise or adapt to increased temperatures: thus, MHWs could pose a greater problem for kelps than background warming [56]. Many kelp species are highly susceptible to MHWs; these events have driven extensive canopy losses [4, 111, 114, 119] and facilitated regime shifts from kelp forests to urchin barrens or turf algae [84, 122, 123]. In extreme cases, MHWs have driven local extinctions of canopy-forming macroalgal species, causing long-term shifts in community composition [86, 117, 122]. The loss of kelp canopies due to ocean warming and MHWs often causes a decline in the abundance and diversity of kelp-associated flora and fauna [4, 67, 111] and can limit the contributions of kelp forests to carbon cycling [96, 125].

Kelp populations from warmer regions may have increased thermal tolerance due to local adaptation compared to populations from cooler regions. Thermal tolerance has two major components, resistance and resilience [97], both of which can be affected by natural selection. Kelp sporophytes and microscopic stages from warmer latitudes often have higher survival rates and fewer indicators of stress at elevated temperatures when compared to cold-temperate populations [71, 79, 80, 94, 95], and have higher maximum temperature thresholds for fertilisation and production of eggs and sporophytes [60, 76, 89], indicating greater resistance. Populations from warmer regions may also recover more quickly after exposure to thermal stress, demonstrating increased resilience [74, 75, 79]. However, kelp populations from cooler regions sometimes display more resistance than populations from warmer regions when exposed to similar temperature anomalies relative to their local mean [5, 23, 35]. This may indicate that the exceedance of species-specific critical temperature thresholds could play a more important role in MHW responses than local adaptation, or that optimal and critical temperature values may not evolve at the same rate [23]. Other physical factors such as upwelling [19, 43] and water depth [2] cause further variability in the physiological responses of kelp populations to extreme temperatures. High genetic connectivity between kelp populations over large spatial scales might also limit their ability to respond to local-scale selection pressures [9]. Heat-tolerant genotypes or lineages have been identified among some kelp species [3, 55, 85, 112], but research into the genetic mechanisms associated with thermal tolerance in macroalgae remains limited and globally inconsistent when compared to other marine taxa [62].

Giant kelp (*Macrocystis pyrifera*) is a globally important foundation species which is highly susceptible to ocean warming and MHWs [4, 20, 67]. *M. pyrifera* is the largest and most widely distributed species of kelp [88, 103] and is among the fastest-growing organisms in the world [22, 100]. *M. pyrifera* is abundant throughout central and southern regions of Aotearoa New Zealand, including most of Te Waipounamu/South Island and Rakiura/Stewart Island, as well as the southern tip of Te Ika-a-Māui/North Island, with its upper thermal limit occurring at around 41°S [57, 105]. Populations rarely persist in regions where sea temperatures exceed 19 °C for months at a time [57], and temperatures of 24 °C and above are lethal to *M. pyrifera* sporophytes [16, 42]. MHWs have contributed to losses of *M. pyrifera* canopy cover throughout New Zealand [114]. Four clusters have been identified for *M. pyrifera* genetic diversity in New Zealand [77]. Genetic divergence is often linked to environmental heterogeneity in *M. pyrifera* [21, 66], so differences in water temperature could feasibly drive changes

in allele frequencies. *M. pyrifera* gametophytes from the Otago region, near the southern edge of the species' range in New Zealand, have a lower maximum temperature threshold for successful fertilisation than those from the more northerly coastal regions of Marlborough and Canterbury [76]. This suggests that local thermal adaptation may have occurred among New Zealand's *M. pyrifera* populations.

It is crucial to understand the variation in thermal tolerance among kelp populations over local and national spatial scales for two reasons. Firstly, it allows for more accurate fine-scale modelling of how these populations might respond to future warming events. Secondly, it could lead to the identification of more susceptible or tolerant populations, which could be prioritised for assisted gene flow or selective breeding within kelp restoration projects [37]. In this study, we subjected sporophytes from five *M. pyrifera* populations spanning the species' latitudinal range in New Zealand (6.5°) to simulated MHW events. We aimed to understand whether these populations differed in their ability to tolerate and recover from MHWs. Based on previous findings on the thermal tolerance of *M. pyrifera* sporophytes in New Zealand [17] and evidence for local thermal adaptation in gametophytes [76], it was hypothesised: 1) that all populations would be negatively affected by temperatures of 18 °C and above, but 2) that the two northernmost populations would experience less severe impacts when compared to the southern populations. This would be investigated through measurements of survivorship, growth, chlorophyll fluorescence, carbon and nitrogen content and stable isotope values. To ensure that we assessed the impacts of adaptation rather than short-term acclimatisation to local temperatures, all sporophytes were cultured at 14 °C for seven months prior to the heatwave experiment.

2 Methods

2.1 Gametophyte collection and culture

Sorus tissue was collected from mature *Macrocystis pyrifera* sporophylls at depths of 3–10 m by diving at five sites throughout New Zealand (Table 1; Fig. 1). The sites spanned a latitudinal gradient of approximately 6.5° (from 40.72°S, 173.97°E to 47.20°S, 167.65°E), with historical summer maximum sea surface temperatures ranging from approximately 14 °C at the southernmost site to 17 °C at the two northernmost sites (baseline climatology; see Supplementary File 1: Fig. S2). The depth disparity between D'Urville Island and the other sites was because

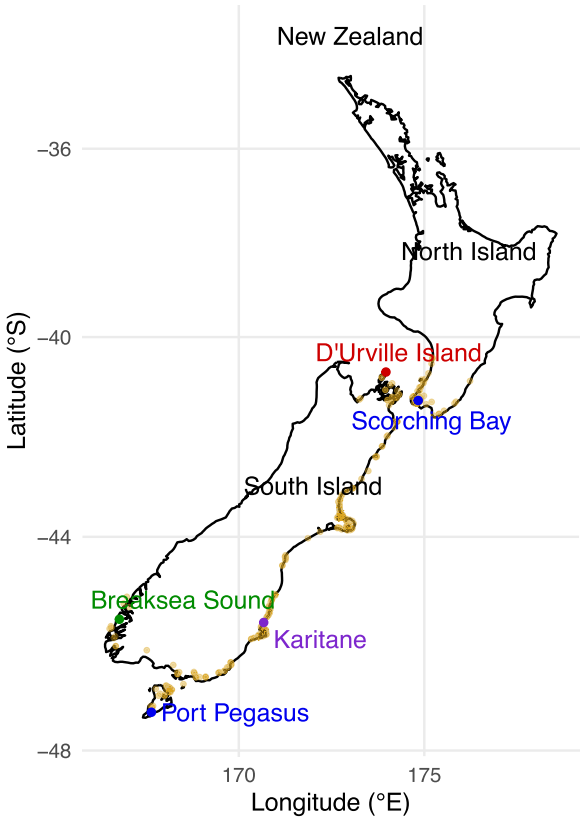


Fig. 1 Locations of *Macrocystis pyrifera* collection sites in Aoteoroa New Zealand, labelled in red, relative to the local distribution of *Macrocystis pyrifera*, shown in brown. *M. pyrifera* distribution data was obtained from GBIF.org [50], using the R package "rgbif" [24], and verified against Hay [57]. Label colours represent distinct genetic clusters

Table 1 Site locations, collection dates, sampling depths, and water temperatures measured when sampling the *Macrocystis pyrifera* sorus tissue used in marine heatwave experiments

Site name	Region	Latitude (°S)	Longitude (°E)	Sampling date (dd/mm/yyyy)	Depth (m)	Temp (°C)
D'Urville Island/Rangitoto ki te Tonga	Marlborough/Te Tau Ihu	40.72	173.97	09/06/2020	10	14
Scorching Bay	Wellington/Te Whanganui-ā-Tara	41.30	174.84	10/10/2020	4	14
Breaksea Sound/Te Puaitaha	Fiordland/Te Rua-o-Te-Moko	45.58	166.78	18/12/2019	3	13
Karitane	Otago/Ōtākou	45.64	170.67	05/03/2020	4	15
Port Pegasus/Pikihatiti	Stewart Island/Rakiura	47.20	167.65	22/09/2020	4	11

this was the only population remaining at this northerly site. The sites also represented the four genetic clusters identified by Le et al. [77]. Up to ten adult sporophytes were sampled from each site; however, at the Breaksea Sound site, the limited availability of healthy adult kelp meant that only one individual could be sampled. Sorus tissue samples were sealed in Ziploc bags full of seawater and transferred in a cooler box in dark conditions to the University of Otago's Portobello Marine Laboratory. It was not part of this study to assess whether the transportation time of parental sorus tissue could have affected offspring fitness. This is acknowledged as a potential variable that was not accounted for. Tissue samples were cleaned gently to remove epiphytes, then wrapped in aluminium foil and refrigerated at 4 °C for 24–48 h before sporulation was induced according to the methodology described in Le et al. [76]. Male and female gametophytes from each population were separated and stored in Eppendorf tubes filled with 0.2 µm filtered seawater (salinity ≈ 32 PSU). These gametophyte cultures were then incubated at 12 °C under red light at an intensity of 5–18 µmol photons m^{-2} s^{-1} to keep the gametophytes in a vegetative state and prevent gamete production as described in Goecke et al. [51]. The samples remained in storage for approximately 3 years before sporophyte production commenced.

Gametophytes were transferred in a cooler to the New Zealand Institute for Earth Science Ltd. (Earth Sciences New Zealand), formerly National Institute of Water & Atmosphere Research Ltd. (NIWA), in July 2022 (transit time < 8 h), where they were again incubated at 12 °C under red light at a salinity of ~ 35 PSU. If a culture showed signs of depigmentation, some gametophytes from that population were transferred to fresh culture medium and temporarily incubated under white light to promote recovery. On 11 July 2023, all gametophytes were fragmented with a pestle and transferred to 1.5 mL Eppendorf tubes filled with ultra-violet (UV)-sterilised seawater enriched with F/2 nutrient solution ([54]; AusAqua, Wallaroo, South Australia) at a salinity of approximately 35 PSU, then incubated at 15 °C under white light to stimulate growth and gamete production. From this point onwards, gametophytes were fragmented again every 1 to 2 weeks. The gametophytes were transferred to fresh F/2 enriched seawater on 26 September 2023.

The long-term storage of kelp gametophytes under red light to delay gametogenesis is a commonly used strategy to create germ banks [51]. This experiment also aimed to assess whether this storage would affect sporophyte viability by comparing these sporophytes to those produced by gametophytes that had not been stored under red light. Sorus tissue was collected from kelp in Scorching Bay in April 2023 and taken directly to Earth Sciences New Zealand's Wellington experimental facility. Spore release was stimulated according to the methods described in Bunting et al. [17], and male and female gametophytes were separated under a microscope. These gametophytes were then cultured in F/2 enriched seawater under white light at 15 °C and fragmented every 1 to 2 weeks to stimulate increased growth. On 9 October 2023, the gametophytes were fragmented once more using a Sunbeam FC7400 food processor (Newell Australia Pty Ltd, Scoresby, Victoria), then males and females were combined and transferred into 600 mL Pyrex jars. The jars contained fresh F/2 seawater and either cotton rope (10 mm in diameter) or 1 mm plastic mesh as a settlement substrate. After observing sporophyte development and growth within these jars for several weeks, it was determined that attachment rates were generally higher on cotton rope, which was then used as the growth substrate for the experiments.

On 31 October 2023, the gametophytes from the experimental populations were fragmented using a Sunbeam FC7400 food processor. The males from each population were combined with the females from the corresponding population, and transferred into 600 mL Pyrex jars containing fresh UV-sterilised F/2-enriched seawater and rope, which were incubated under white light at 15 °C. The medium in the jars was replaced at least once a week, and cultures were observed under a microscope weekly to check for developing sporophytes. The sporophytes from the two Scorching Bay cultures were compared to test whether storing the gametophytes under red light had any long-term effects on sporophyte survival and physiology.

2.2 Sporophyte culture and experimental setup

Most of the cultures contained sporophytes visible to the naked eye by 28 February 2024. The culture ropes were transferred to experimental tanks in Earth Science New Zealand's Marine Environmental Manipulation Facility on this date. Each population was kept in a separate tank throughout the culture period. The setup comprised 12 × 5 L capacity tanks (176 × 325 × 150 mm), each connected to a separate 20 L sump (i.e., $n = 4$ per treatment). The tanks were located in a temperature-controlled room set to a base temperature of 12 °C and kept under a 12:12 light:dark cycle. Light was provided by 4Seasons 100 W Quantum PAR LED light panels (4Seasons, Auckland, New Zealand) at a mean intensity of 56 µmol of photosynthetically active radiation (PAR) m^{-2} s^{-1}, giving a total daily dose of approximately 2.42 mol PAR m^{-2} d^{-1}. The sumps were filled with seawater from Wellington Harbour (salinity ≈35 PSU) that had been pumped through a 30 W UV steriliser (Trevoli, Auckland, New Zealand).

The temperature in each sump was individually controlled by modified EHEIM Thermocontrol 150 W heating elements (EHEIM, Deizisau, Germany). Water was pumped from each sump into the corresponding tank by a Hailea BT700B submersible pump (flow rate 690 L h^{-1}; Hailea, Guangdong, China) attached to a 12 mm diameter hose, then flowed back into the sump through another hose. The temperature of the tanks was maintained at 14 ± 1 °C throughout the culture period.

The tanks were set up as a recirculating system initially, from 28 February 2024. Sumps were partially drained and refilled with fresh UV-sterilised seawater twice a week. Nutrients were supplied by adding 4 mL of AlgaBoost F/2 1000× concentrate (AusAqua, Wallaroo, South Australia) to each tank, three times a week. The tank setup was modified into a flow-through system on 29 April 2024 to create ideal growing conditions (as per [17]). Each tank was connected to one of six 55 L header tanks containing UV-sterilised seawater, with two tanks connected to each header. Seawater was pumped out of the header tanks and into the experimental tanks at a rate of approximately 120 mL min^{-1}. An upstand with an outflow hose and a dump valve was added to each sump to prevent overfilling. A full timeline of the collection and culture of gametophytes and sporophytes is provided in Supplementary File 1, Table S1.

To ensure consistency, temperature, pH, dissolved oxygen concentration, and salinity were measured weekly in the experimental tanks throughout the culture period and heatwave experiment. Temperature was measured using a CENTER 376 Datalogger Precision RTD Thermometer (Center Technology Corp, Taipei, Taiwan); pH, dissolved oxygen and salinity were measured using Hach Intellical probes (PHC101, LDO101, and CDC401, respectively; Hach New Zealand, Auckland, New Zealand). The measured values throughout the experimental period are summarised in Supplementary File 1, Table S2. Probes were calibrated weekly against known standards. The pH electrode was calibrated against both Tris buffer in artificial seawater [34] and Orion pH buffers (ThermoFisher Scientific, Waltham, MA, USA); tank pH was measured potentiometrically in millivolts, then converted to the total scale using the R package "seacarb" [47]. The dissolved oxygen probe was calibrated against a 0% standard (water with anhydrous sodium sulphite added) and a 100% standard (water aerated by an air stone,Aqua One, Ingleburn, Australia), whilst the salinity probe was calibrated against 1000 µS cm^{-1} sodium chloride standard solution (Hach, Loveland, CO, USA). In addition, HOBO Pendant MX2202 loggers (Onset, Bourne, MA, USA) were placed in the sumps for long-term temperature monitoring. Seawater samples for chemical analysis were collected every second week during the heatwave and recovery phases, based on methods derived from Pritchard et al. [98]. A water sample was taken from each experimental tank. Samples were collected using a 50 mL syringe and passed through a 0.45 mm filter into a 50 mL plastic vial, then frozen at − 20 °C. A separate syringe and filter were used for each sample. The water samples were analysed subsequently at the University of Otago's Portobello Marine Laboratory. A Lachat QuikChem 8500 Series 2 FIA auto analyser (Hach New Zealand, Auckland, New Zealand) was used to calculate the concentrations of NO_x ions, ammonia, and orthophosphate using methods derived from Strickland and Parsons [113]. The nitrogen and phosphate availability data are provided in Supplementary File 1: Table S3.

2.3 Heatwave experiment

Twenty-four macroscopic sporophytes (blade length ≥ 4 mm) were selected from each of the cultured populations except for the D'Urville Island culture, where only 12 sporophytes were selected due to a lower abundance of large sporophytes. The number and average day 0 size of sporophytes taken from each culture are summarised in Supplementary File 1: Table S4. These sporophytes were attached to separate, labelled pieces of mesh, which were weighed down with sinkers attached to plastic pegs to keep the sporophytes submerged. One sporophyte from D'Urville Island and two sporophytes from each of the other cultures were randomly assigned to each of the 12 experimental tanks. Sporophytes were left for a week to reattach before temperature ramping commenced.

The heatwave experiment began on 5 June 2024. Two heatwave treatments were used, an 18 °C heatwave and a 22 °C heatwave, as well as a control in which temperatures remained at 14 °C. The temperature treatments were interspersed in a randomised block design (see A-2 in [63]) with three tanks in each block, and each treatment occurring once in each block. At the beginning of the heatwave simulation, tank temperatures were raised by increments of 2 °C per day. The temperatures were lowered to 14 °C in 2 °C daily increments at the end of the heatwave period. The total duration of the heatwave period, including temperature ramping, was 22 days. After the conclusion of the heatwave period, the sporophytes remained in the tanks for a further 21 days at 14 °C to recover. A detailed timeline of the heatwave experiment, including dates on which measurements and samples were taken, is provided in Supplementary File 1: Fig. S1.

To compare these heatwave treatments to real-world MHW data from the regions from which the parental gametophytes were collected, the National Oceanic and Atmospheric Administration's "Optimum Interpolation Sea Surface Temperature V2.1" dataset [101] was used

to obtain daily sea surface temperature data for the five regions of New Zealand from which kelp cultures were sourced (Table 1). These data were used to construct a baseline seasonal climatology from 1982 to 2011 for each region to detect and analyse MHWs that occurred between 1982 and 2023, using the R package "heatwaveR" [104]. The control temperature of 14 °C is close to the historical summer maximum for the Otago and Stewart Island regions and is near the middle of the typical yearly temperature range for the other regions studied (Supplementary File 1: Fig. S2). Since 1990, all these regions have experienced several MHW events which exceeded 22 days in duration (Supplementary File 1: Fig. S3), and at least one event in which absolute temperatures reached or exceeded 18 °C (Supplementary File 1: Fig. S2). As of 2024, none of these regions have recorded temperatures of 22 °C, or temperature anomalies of 8 °C, during MHW events (Supplementary File 1: Fig. S2, S4); this treatment represents a hypothetical future (late twenty-first century) severe MHW under a high greenhouse gas emissions scenario (SSP5-8.5; [91]). Under SSP5-8.5, mean MHW intensities within New Zealand are projected to increase by up to 2 °C by 2100 [10], so mean temperature anomalies within the study regions could approach 4 °C, whilst anomalies of 8 °C would probably remain rare but become more likely.

2.4 Measurements

2.4.1 Survivorship

Survivorship was assessed at the conclusion of the heatwave and the recovery period. Sporophytes were marked as dead if the blade eroded completely, or if the holdfast detached from the mesh. The sporophytes that were sacrificed for chemical analysis after the heatwave were excluded from the survivorship data for the recovery period.

2.4.2 Growth

Growth measurements were taken on the day before the heatwave simulation started, the second-to-last day of the heatwave, and the second-to-last day of the recovery period. Length was measured to the nearest millimetre using a ruler. Wet weight was measured using a Scout SPX2202 fine balance (OHAUS Corporation, Parsippany, NJ, USA) with a precision of ± 5 mg. Pegs and sinkers were removed during weight measurements. Relative growth rates (RGR; [69]) were calculated for the heatwave and recovery periods.

2.4.3 Chlorophyll fluorescence

A Diving-PAM blue light fluorometer (Walz, Effeltrich, Germany) was used to measure the maximum quantum yield of photochemical conversion (ratio of variable fluorescence F_v to maximum fluorescence F_m) of the sporophyte blades. Fluorometry settings were optimised as per Enríquez and Borowitzka [39], the intensity of the saturating pulse was 5500 µmol photons m^{-2} s^{-1}, and the duration was 0.8 s. Measurements were taken 5 days prior to the heatwave, on the final day of the heatwave, and on the final day of the recovery period. Before taking measurements, sporophytes were kept in darkness for 30 min to ensure dark adaptation of photosystem II [83].

2.4.4 Total % carbon, % nitrogen, and stable isotope values

At the conclusion of the heatwave experiment, tissue samples for carbon and nitrogen stable isotope analysis were taken from roughly 30% of the surviving sporophytes from the control and 18 °C treatments. The 22 °C treatment was not destructively sampled at this timepoint due to low survival rates, although one blade that broke off during growth measurements was kept aside for analysis. After the conclusion of the recovery period, samples were taken from all surviving sporophytes of sufficient size (blade length > 5 mm), with the whole blade being used in most cases. The total number of individuals from each population sampled at each timepoint is given in Supplementary File 1: Table S5. Samples were taken with a razor blade which was rinsed with ethanol and water between samples to prevent cross-contamination. Tissue samples were dried in an Isotherm convection oven (Esco Lifesciences Group, Singapore) at 75 °C for at least 24 h, then homogenised by grinding with a mortar and pestle and sealed in Eppendorf® tubes.

Dried tissue samples were analysed at Earth Science New Zealand's Environmental and Ecological Stable Isotope Analytical Facility in Wellington. Larger samples were divided into two subsamples before analysis. Stable isotope analyses were carried out on a Delta V Plus continuous flow isotope ratio mass spectrometer linked via a ConFlo IV to a Flash 2000 elemental analyser (Thermo-Fisher Scientific, Bremen, Germany) using either a ZeroBlank autosampler (Costech International, Milan, Italy) for low-weight samples, or a MAS 200 R autosampler (Thermo-Fisher Scientific, Bremen, Germany) for larger samples. For low-weight samples and standards, the Flash 2000 EA was configured in low volume set-up (IVA Analysentechnik, Meerbusch, Germany). Reference materials were used to determine stable isotope values following Paul et al. [92]. Sample $\delta^{15}N$ values were two-point normalised using stable isotopic data from the daily analysis of NIST 8573 USGS40 L-glutamic acid and NIST 8548 IAEA-N2 Ammonium sulphate. Sample $\delta^{13}C$ values were two-point normalised using stable isotopic data from the daily analysis of NIST 8573 USGS40 and L-Valine #2 USGS74. Precision was determined by the

repeat analysis of the working laboratory standards DL-Leucine (DL-2-Amino-4-methylpentanoic acid, C6H13NO2, Lot 127H1084, Sigma, Australia), IPE4-4 and Kaolinite. Data from the analysis of USGS65 Glycine (values of both $\delta^{15}N$ and $\delta^{13}C$) and L-Valine #2 USGS74 (value of $\delta^{15}N$ only) were used to check accuracy and precision. Repeat analysis of these standards produced data accurate to within 0.2%, with a precision of better than 0.2% for both %N and %C content and data accurate to within 0.2 ‰, with a precision of better than 0.2 ‰ for both $\delta^{15}N$ and $\delta^{13}C$. For additional analytical details, see Bury et al. [18], (Supplementary Materials, Text S6).

2.5 Statistical analysis

The R package "survival" [116] was used to run Kaplan–Meier survival analysis [70] on the cumulative survival rates of sporophytes at the end of the heatwave and recovery periods, assessing the impacts of temperature, population, and their interaction on survivorship. The sporophytes that were sacrificed for chemical analysis after the heatwave were excluded from survival analysis during the recovery phase. A chi-squared test for independence [93] was run on the survivorship data to confirm that the effects of temperature and population were independent. After using the Shapiro–Wilk test [106] to confirm that these datasets were normally distributed, the R package "car" [46] was used to carry out Levene's test for equality of variances [78] and confirm that there were no violations of homoscedasticity. The R package "lme4" [8] was then used to fit linear mixed-effects models to the data for growth, chlorophyll fluorescence, total carbon and nitrogen content, and stable isotope values. These models used temperature, population, and their interaction as fixed effects, with header tank used as a random effect. The R package "car" [46] was used to run analysis of variance (ANOVA) on these models. If significant effects were found, the R package "multcomp" [61] was used to run Tukey's pairwise tests [120] on these models and identify any significant differences between individual treatments.

3 Results

Temperature, pH, salinity, dissolved oxygen, nitrogen, and orthophosphate data for the duration of the heatwave and recovery periods are summarised in Supplementary File 1: Tables S2–3. Tank temperatures consistently remained within 1 °C of target values. The pH and dissolved oxygen values and chemical content fluctuated slightly over the course of the experiment, but remained similar across tanks at each timepoint.

3.1 Survival

In total, 68% of all *Macrocystis pyrifera* sporophytes survived the heatwave period. The 22 °C heatwave treatment had the highest mortality rates, with 36% of sporophytes surviving, compared to 77% in the 18 °C heatwave treatment and 91% in the control. This trend was consistent across all populations, although the difference in survival rates between the control and 22 °C treatments was less pronounced for the two northernmost cultures, Scorching Bay and D'Urville Island, as these northern populations tended to have slightly lower survival rates than the southern populations at 14 °C and 18 °C (Fig. 2a). Conversely, the Scorching Bay culture that was not stored under red light had the lowest survival rate of any culture at 22 °C, with just one sporophyte surviving the heatwave (Scorching Bay A, Fig. 2a). Temperature had a strongly significant impact on survival rates ($p < 0.001$, Table 2), and an interactive effect of temperature and population was also detected ($p < 0.001$, Table 2), potentially caused by the high mortality among the Scorching Bay A culture at 22 °C; population alone did not have a significant effect ($p = 0.600$, Table 2).

Most of the surviving sporophytes from the 22 °C treatment died by the end of the recovery phase, including all 16 sporophytes from the two Scorching Bay cultures and all four from D'Urville Island (Fig. 2b). Conversely, most of the remaining sporophytes in the other temperature treatments survived the recovery phase: 96% survived in the control, and 88% in the 18 °C treatment. Survival rates at 14 °C and 18 °C were slightly lower among the northern cultures. Cumulative survival rates for the whole experimental period were 84% in the control, 62% in the 18 °C treatment, and 9% in the 22 °C treatment, and 48% overall (Fig. 2b). Temperature and the interaction between temperature and population both had strongly significant effects ($p < 0.001$) on cumulative survival rates throughout the experiment (Table 2). The initial size of sporophytes did not seem to affect their probability of survival (Supplementary File 1: Fig. S5).

3.2 Growth

The sporophytes within the 22 °C heatwave treatment had much lower mean linear RGR ($-1.41 \pm 0.41\%$ day^{-1}; mean ± standard error) than those in the 14 °C control ($2.43 \pm 0.52\%$ day^{-1}); this trend was consistent across all populations, even though growth rates varied within and between cultures (Fig. 3a). Conversely, there was no consistent difference in mean RGR between the 18 °C heatwave treatment ($2.02 \pm 0.55\%$ day^{-1}) and the control. The two southernmost cultures, Karitane and Port Pegasus, tended to have higher mean RGR than many of the other cultures across all temperature treatments (Fig. 3a). Both

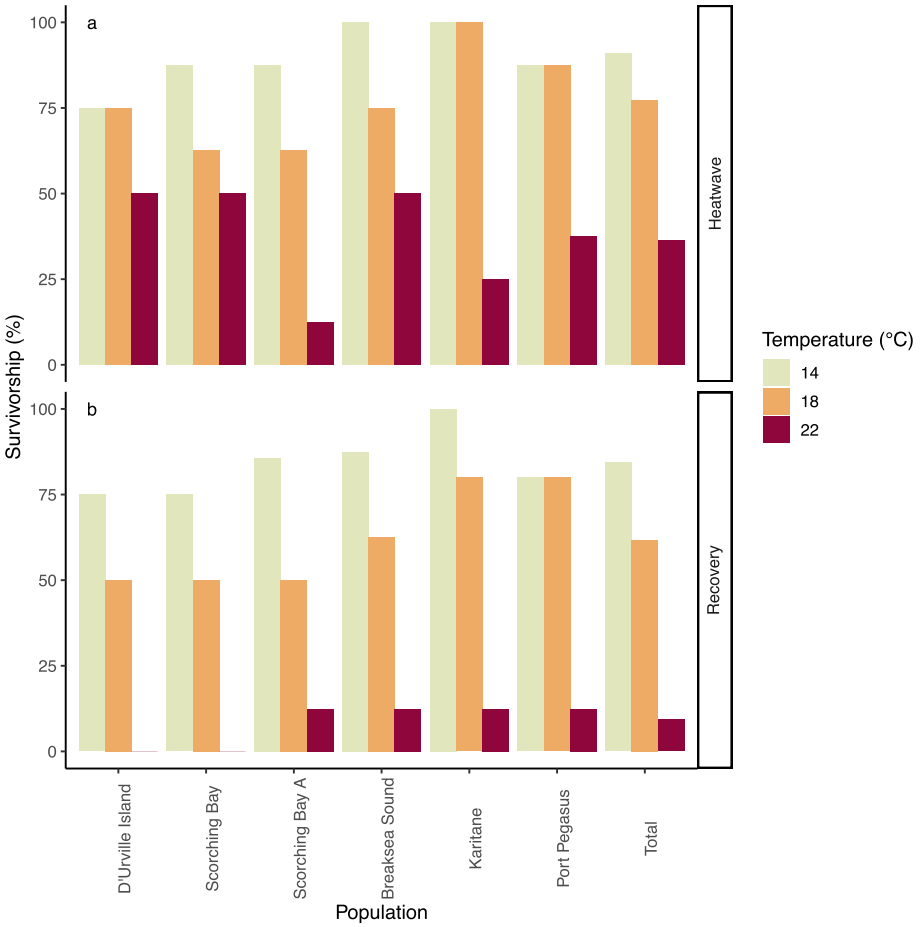


Fig. 2 Cumulative survivorship rates of *Macrocystis pyrifera* sporophytes in each temperature treatment at the conclusion of: **a** the heatwave phase and, **b** the recovery phase of the experiment. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

Table 2 Summary of *p*-values obtained from estimating the effects of heatwave temperature (p_T), population (p_P), and their interaction (p_{T*P}) on cumulative survival rates, relative growth rates (RGR), maximum quantum yield (F_v/F_m), carbon and nitrogen content, carbon: nitrogen (C:N) mass ratio, and stable isotope values in *Macrocystis pyrifera* sporophytes during the heatwave period and recovery phase

	Heatwave				**Recovery**			
	R_T	p_T	p_P	p_{T*P}	R_T	p_T	p_P	p_{T*P}
Survival		**< 0.001**	0.600	**< 0.001**		**< 0.001**	0.800	**< 0.001**
Linear RGR	**− 0.428**	**< 0.001**	**< 0.001**	0.894	− 0.063	0.935	0.089	0.305
RGR by weight	− 0.123	0.679	0.642	0.368	+ 0.226	0.683	0.063	0.318
F_v/F_m	− 0.002	0.082	0.278	0.383	− 0.009	0.204	0.527	0.774
Total carbon (%)	**+ 0.471**	**< 0.001**	0.477	0.805	− 0.303	0.467	0.930	0.042
Total nitrogen (%)	+ 0.015	0.303	0.181	0.862	+ 0.001	0.586	0.224	0.531
C:N mass ratio	**+ 0.452**	**< 0.001**	0.381	0.874	− 0.171	0.605	**0.044**	0.076
$\delta^{13}C$ (‰)	+ 0.352	0.079	0.225	0.851	+ 0.321	0.893	0.363	0.658
$\delta^{15}N$ (‰)	**− 0.103**	**0.002**	**0.021**	**0.002**	**+ 0.518**	**0.047**	0.725	0.676

The direction and strength of the relationship between temperature and *M. pyrifera* physiological parameters, i.e. the predicted response, R_T, of each parameter under a 1 °C increase in temperature, is also provided where linear mixed-effect models were applied to the data. Header tank was treated as a random effect. Statistically significant effects ($p < 0.05$) are indicated in bold

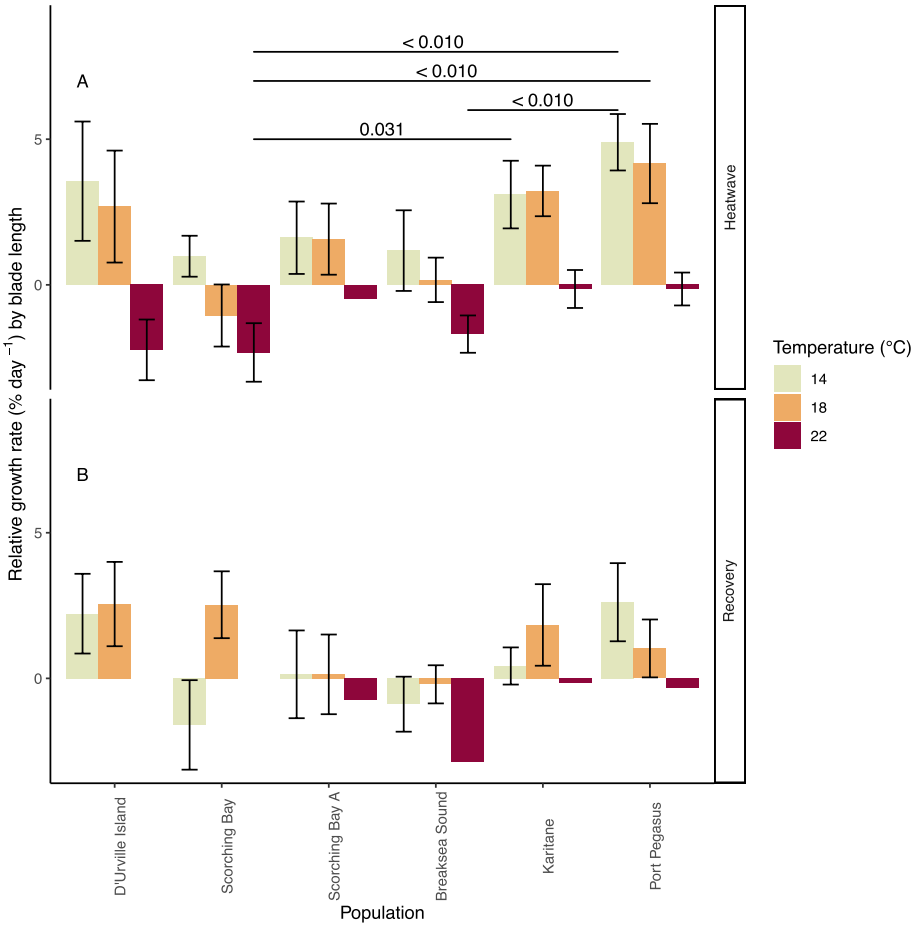


Fig. 3 Mean relative growth rates (RGR), with standard error ($n = 8$), obtained from blade length of *Macrocystis pyrifera* sporophytes in each temperature treatment throughout: **a** the heatwave phase and, **b** the recovery phase of the experiment. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

temperature ($p < 0.001$, Table 2) and population ($p < 0.001$, Table 2) had strongly significant effects on linear RGR during the heatwave; there was no evidence for an interaction between the two variables ($p = 0.894$, Table 2). When comparing individual treatment-population combinations, the sporophytes from the Port Pegasus culture within the control had a significantly higher mean RGR than the Breaksea Sound sporophytes within the 22 °C treatment ($p < 0.010$) and the Scorching Bay sporophytes within both the 18 °C and 22 °C treatments ($p < 0.010$ and $p = 0.031$, respectively).

Population did not have a significant effect on linear RGR during the recovery phase (though the effect was near-significant: $p = 0.089$, Table 2), and mean RGR were generally lowest in the Scorching Bay and Breaksea Sound cultures (Fig. 3b). Temperature did not have a significant impact overall on linear RGR during the recovery phase ($p = 0.935$, Table 2), although all the surviving sporophytes that had been exposed to the 22 °C treatment had negative RGR, with a mean RGR of $-1.02 \pm 0.62\%$ day^{-1}, whereas RGR were more variable among the 14 °C and 18 °C treatments (Fig. 3b). There was no interactive effect of temperature and population on linear RGR ($p = 0.305$, Table 2).

There were no consistent trends in RGR by weight, as the level of variability within populations and treatments tended to be much greater than any differences between treatments (Fig. 4). Neither temperature, population, nor their interaction had any statistically significant impacts on weight-based RGR during either the heatwave or the recovery phase (Table 2), although population had a near-significant impact during the recovery phase ($p = 0.063$). The only significant difference found between individual treatment-population combinations was between the Breaksea Sound and Port Pegasus cultures within the control during the recovery phase, with Port Pegasus having significantly higher RGR ($p = 0.028$).

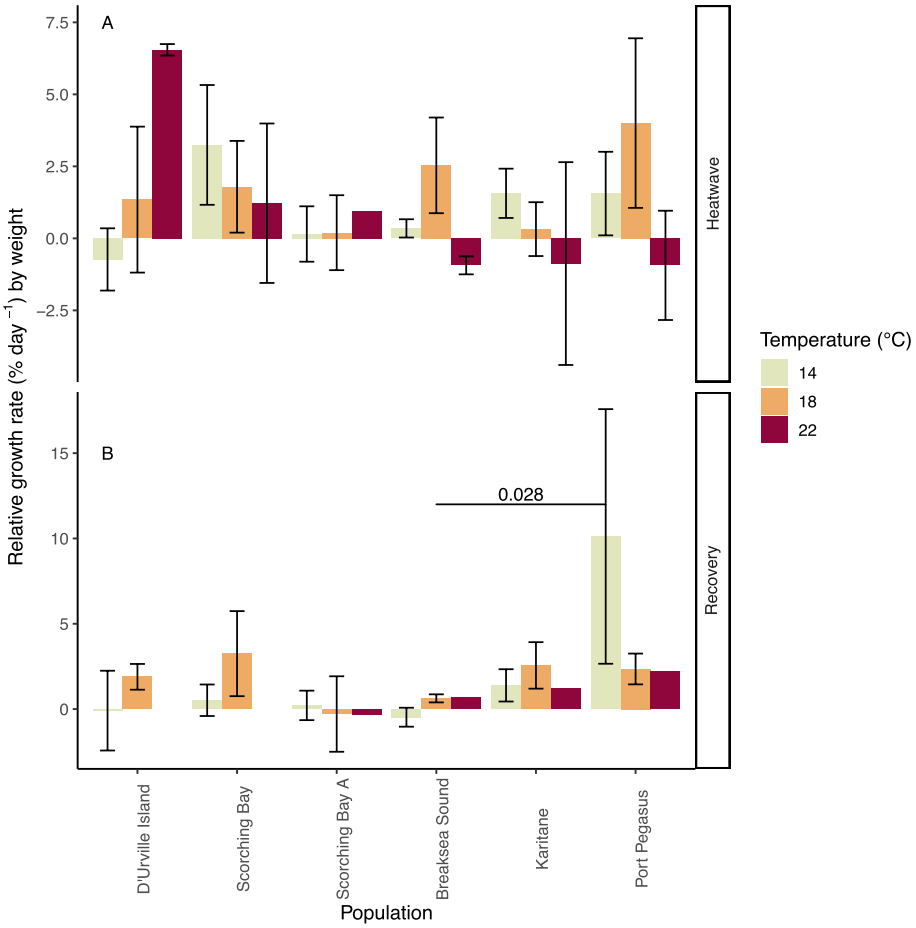


Fig. 4 Mean relative growth rates (RGR), with standard error ($n = 8$), obtained from wet weight of *Macrocystis pyrifera* sporophytes in each temperature treatment throughout: **a** the heatwave phase and, **b** the recovery phase of the experiment. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

3.3 Chlorophyll fluorescence

There were no consistent differences in maximum quantum yield (F_v/F_m) values between populations or tanks prior to the heatwave (Fig. 5a), with an overall mean F_v/F_m value of 0.688 ± 0.006. After 3 weeks in heatwave conditions, mean F_v/F_m values were lower at 22 °C than at 14 °C or 18 °C for every culture, except Karitane and the one surviving sporophyte in the non-red-light-exposed Scorching Bay culture (Fig. 5b); this difference in F_v/F_m values was near-significant ($p = 0.082$, Table 2). The overall mean F_v/F_m value was 0.669 ± 0.011 at 22 °C, compared to 0.703 ± 0.006 in the control and 0.700 ± 0.011 at 18 °C. Population had no significant effect on F_v/F_m ($p = 0.278$, Table 2) during the heatwave, and there was no interactive effect of temperature and population ($p = 0.383$, Table 2).

At the end of the recovery phase, the surviving sporophytes from the 22 °C treatments had a relatively low mean F_v/F_m value (0.645 ± 0.010) when compared to the control (0.684 ± 0.008) and the 18 °C treatment (0.678 ± 0.011) (Fig. 5c). However, temperature did not have a statistically significant effect on F_v/F_m during the recovery phase ($p = 0.204$, Table 2), and neither did population ($p = 0.527$, Table 2) nor the interaction between the two variables ($p = 0.774$, Table 2).

3.4 Total % carbon, % nitrogen, and stable isotope values

At the end of the heatwave, the sporophytes within the 18 °C treatment had significantly higher mean total carbon content (22.5 ± 0.3%) than the control (20.2 ± 0.5%); this trend was consistent across all populations sampled (Fig. 6a; $p < 0.001$, Table 2). The one sporophyte analysed from the Scorching Bay culture at 22 °C also had very high total carbon content (24.1 ± 0.86%, based on two subsamples). However, there was no strong relationship between temperature and carbon content in the samples taken at the end of the recovery period (Fig. 6b; $p = 0.467$, Table 2), although there was a significant interactive

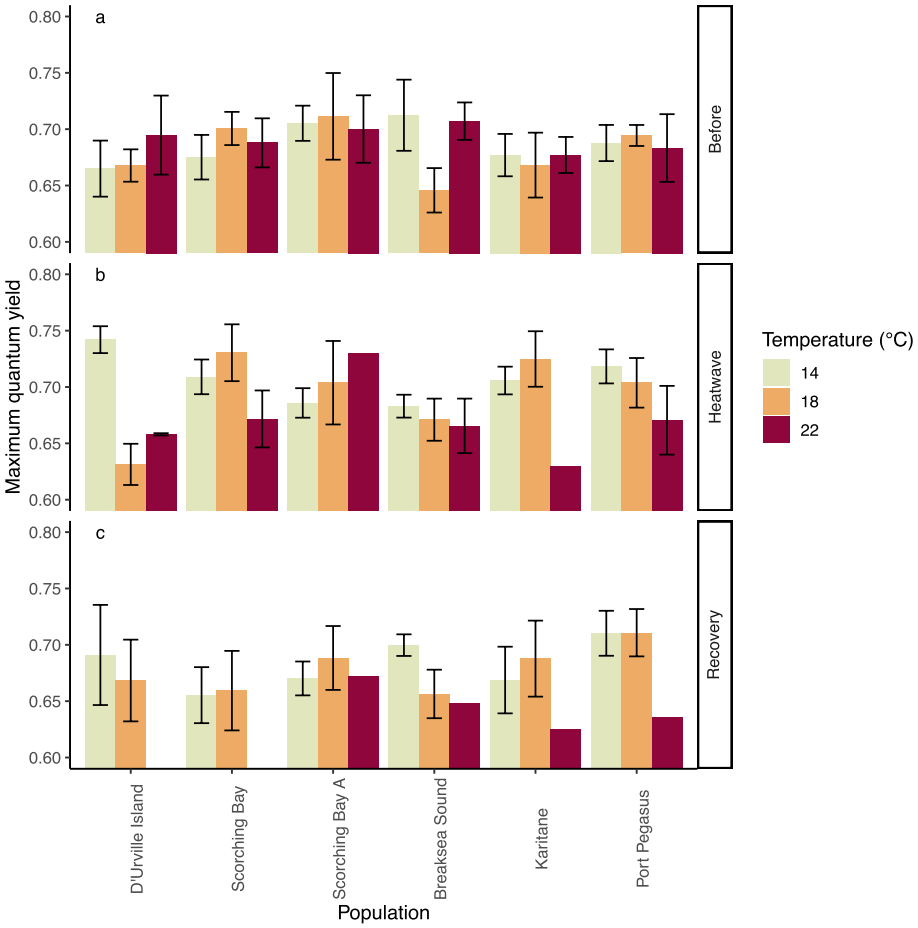


Fig. 5 Mean maximum quantum yield of photochemical energy conversion (F_v/F_m) values, with standard error ($n=8$), of *Macrocystis pyrifera* sporophytes in each temperature treatment, measured: **a** prior to the heatwave experiment, **b** at the conclusion of the heatwave phase, and **c** at the conclusion of the recovery phase. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

effect of temperature and population ($p=0.042$). Neither temperature nor population had a consistent impact on total nitrogen content (Table 2; Fig. 7). Consequentially, the carbon to nitrogen (C:N) mass ratio was also higher at 18 °C (11.6 ± 0.4) than in the control (10.7 ± 0.5) during the heatwave (Fig. 8a; $p<0.001$, Table 2), whilst the sporophyte analysed from the 22 °C treatment had a C:N mass ratio of 15.8 ± 0.9, which was significantly higher than most of the cultures kept at 14 °C or 18 °C (Fig. 8a). At the end of the recovery phase, no relationship was found between temperature and C:N mass ratio ($p=0.493$, Table 2), however, there was a significant effect of population ($p=0.044$), as well as a near-significant interactive effect ($p=0.076$), with the sporophytes from the Karitane culture kept at 14 °C and 18 °C having particularly low C:N mass ratios (10.3 ± 0.3 at 14 °C and 9.85 for the one surviving sporophyte at 18 °C; Fig. 8b).

Sporophyte $\delta^{13}C$ values tended to increase with increasing temperature among the samples taken immediately after the simulated heatwave (Fig. 9a), with mean values of − 29.9 ± 0.6 ‰ in the control (14 °C), compared to − 28.5 ± 0.7 ‰ at 18 °C and − 27.7 ± 0.6 ‰ for the sporophyte sampled from the 22 °C treatment. The relationship between temperature and $\delta^{13}C$ values was near-significant ($p=0.079$, Table 2), but no such trend was observed after the recovery phase (Fig. 9b). Population also had no independent or interactive effect on $\delta^{13}C$ values during the heatwave or the recovery phase (Table 2). Temperature had a significant positive relationship with sporophyte $\delta^{15}N$ values during the simulated heatwave (Fig. 10a; $p=0.002$, Table 2), with mean values of 1.11 ± 0.34 ‰ in the control (14 °C), 3.03 ± 0.58 ‰ at 18 °C, and 2.67 ± 0.51 ‰ at 22 °C. Population ($p=0.021$, Table 2) and the interaction between temperature and population ($p=0.002$, Table 2) also had significant impacts on $\delta^{15}N$ values during the heatwave, with the Scorching Bay A and Port Pegasus cultures having higher mean $\delta^{15}N$ values at 18 °C (4.27 ± 0.31 ‰ and 4.05 ± 1.05

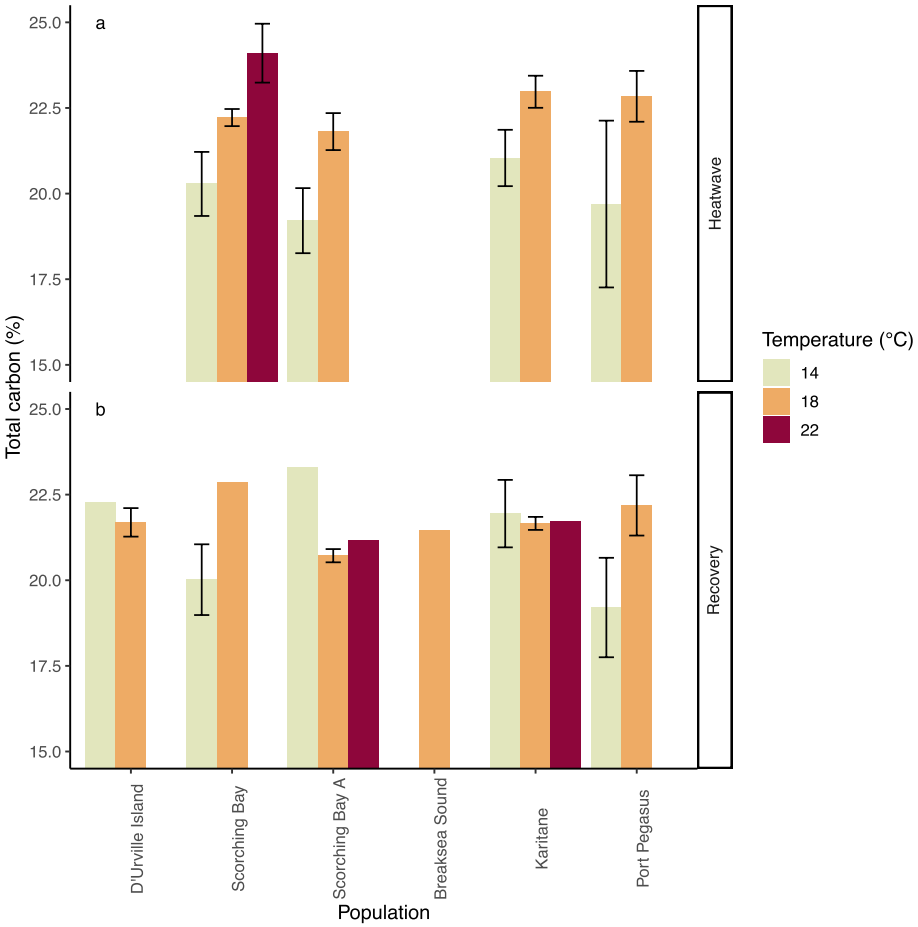


Fig. 6 Mean total carbon content, with standard error (see Table S5 for *n*) of *Macrocystis pyrifera* blade tissue samples taken: **a** immediately after the heatwave and, **b** at the conclusion of the recovery phase. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

‰ respectively; Fig. 10a). Temperature also had a significant positive effect on $\delta^{15}N$ values at the end of the recovery phase ($p = 0.047$, Table 2), whilst population had no significant individual or interactive effect at that timepoint (Table 2; Fig. 10b).

4 Discussion

Juvenile *Macrocystis pyrifera* sporophytes from five populations spanning their natural latitudinal range in New Zealand were negatively affected by a simulated MHW at 22 °C, but comparatively resistant to an 18 °C MHW. The initial hypothesis that the northern cultures would be more heat-tolerant was not supported; cumulative survival rates after the heatwave experiment were slightly lower among the Wellington and Marlborough cultures, and the impacts of heatwaves on growth and chemical composition were generally similar across all populations studied, with few interactive effects of temperature and population. Therefore, the study found no clear evidence that local adaptation to high temperatures has occurred among sporophytes from northerly *M. pyrifera* populations in New Zealand. The sporophytes used in the experiment represent between one and ten parental genotypes per site, which may not represent the full genetic diversity of each population, so these findings should be interpreted with some caution. Additionally, high mortality rates mean that there is limited replication of some temperature-population combinations, and the analysis of multiple endpoints without adjustment could increase the risk of a Type I error, i.e. erroneously interpreting results as significant.

There is evidence that local thermal adaptation has occurred among *M. pyrifera* gametophytes in New Zealand, with cultures from the Marlborough coastal region having a maximum temperature threshold for successful fertilisation that was approximately 3 °C higher than cultures from the Otago coastline [76], although we found no clear evidence for similar adaptation in sporophytes.

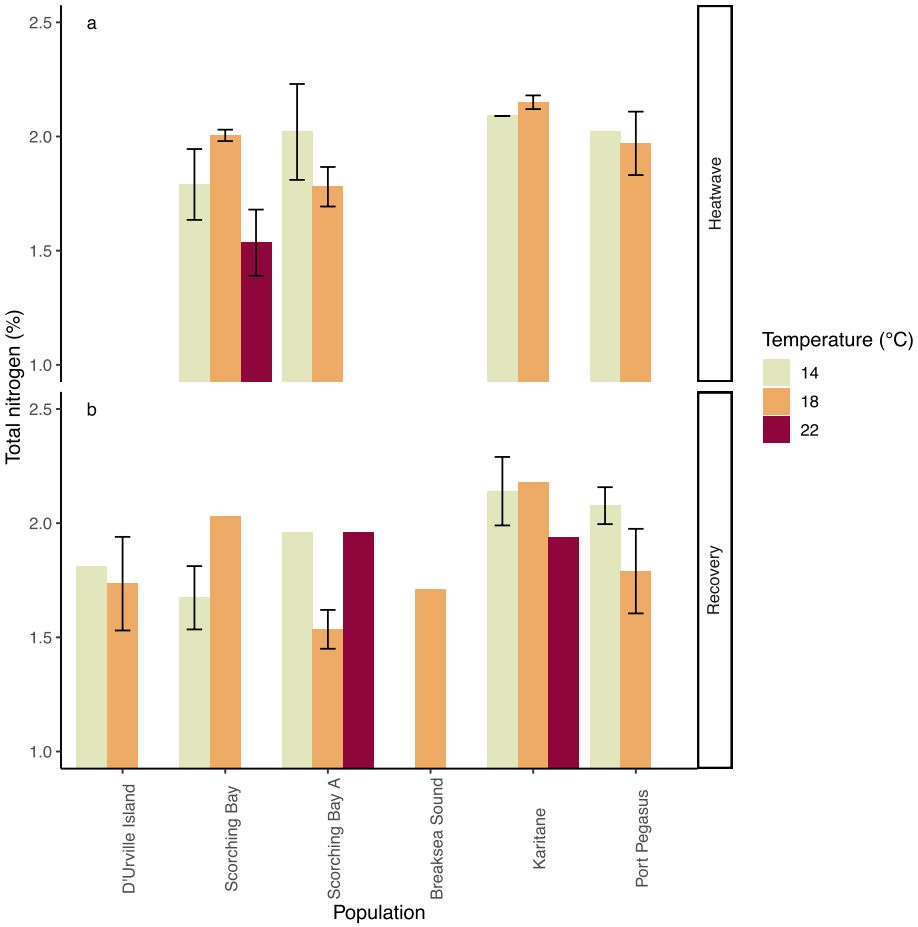


Fig. 7 Mean total nitrogen content, with standard error (see Table S5 for *n*), of *Macrocystis pyrifera* blade tissue samples taken: **a** immediately after the heatwave and, **b** at the conclusion of the recovery phase. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

M. pyrifera gametophytes have a higher thermal tolerance range than sporophytes [74]. It has been suggested that this discrepancy might reduce the species' capacity to adapt to extreme temperatures, as gametophytes might be resistant to temperatures that cause tissue damage in sporophytes [9]. Kelp gametophytes can delay sporophyte production under sub-optimal environmental conditions [36], so if few sporophytes are released during extreme warming events, selection pressure due to temperature may be minimised. The mismatch in thermal responses between *M. pyrifera* sporophytes and gametophytes suggests that researchers must evaluate the effects of temperature on all life history stages of kelp to accurately model how local kelp populations will respond to future climate change and MHWs.

Rather than population-level adaptation, the thermal responses of the sporophytes in this experiment might have been more strongly affected by the temperature at which they were cultured. *M. pyrifera* sporophytes have high phenotypic plasticity, with their morphology varying greatly in response to environmental conditions [31]. The sporophytes used in this experiment were maintained at 14–15 °C over a total period of approximately 7 months, and parental gametophytes were kept at this temperature for a further 2 months, whereas wild kelp in Karitane and Port Pegasus would typically only experience this temperature range for a brief period during summer (Supplementary File 1: Fig. S2d–e). Kelp species' responses to MHWs can vary greatly between seasons, and it is thought that both cumulative stress and thermal acclimatisation may drive this variability [40, 90]. It has been suggested that "thermal priming", culturing organisms at a temperature that is warmer or colder than they would frequently encounter in the wild, could induce epigenetic changes that cause organisms to develop a "stress memory" that allows them to respond more effectively to future thermal stress [68]. There is evidence that warm thermal priming of early life-history stages can improve

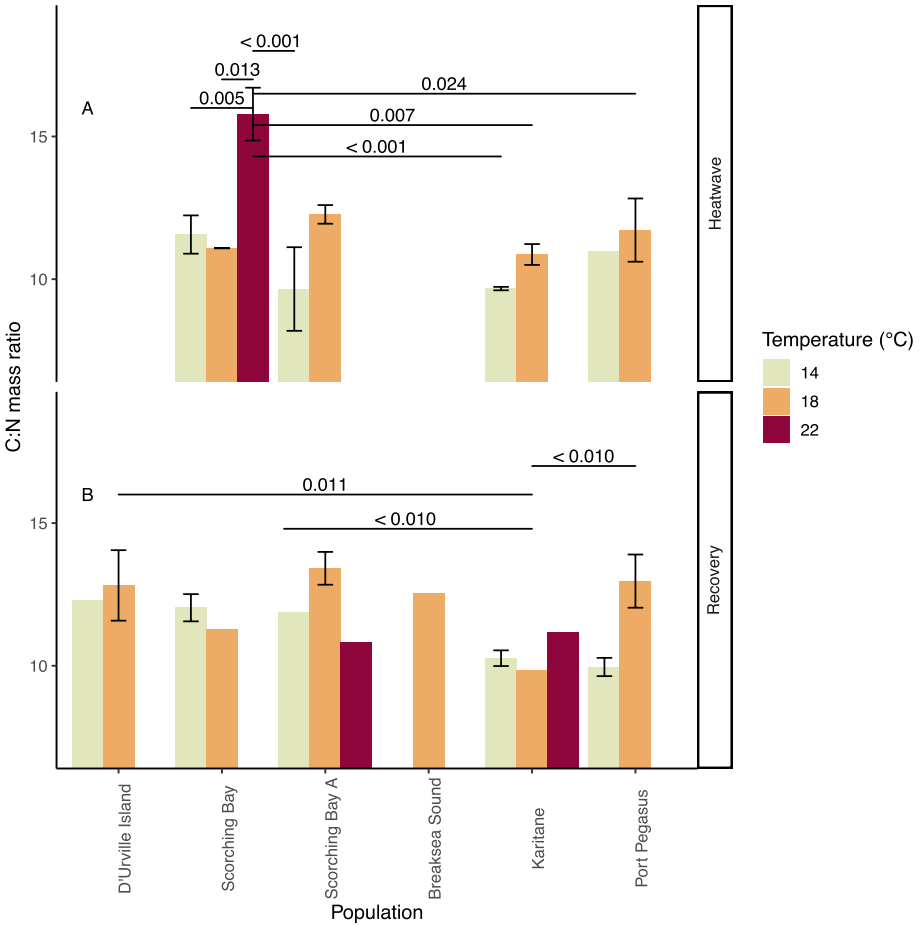


Fig. 8 Mean carbon: nitrogen (C:N) mass ratio, with standard error (see Table S5 for *n*), of *Macrocystis pyrifera* blade tissue samples taken: **a** immediately after the heatwave and, **b** at the conclusion of the recovery phase. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

thermal tolerance in kelps ([Wu, 1998, as cited in 13, 48, 99]). Hence, the fact that the sporophytes studied here were cultured at identical temperatures might have led them to develop similar thermal tolerance ranges through physiological acclimatisation.

The results of this study concur with data from other countries showing that *M. pyrifera* populations do not always adapt to local temperatures. In Chile, whilst there are differences in thermal tolerance between *M. pyrifera* populations, the northernmost populations are not necessarily the most tolerant [9, 19]. Although laboratory studies have found evidence for local thermal adaptation in *M. pyrifera* gametophytes and sporophytes in North America [60, 74, 75], field evidence suggests that the exceedance of critical thermal thresholds throughout the species' whole range during MHWs can swamp the effects of variation between populations [23]. High genetic connectivity between geographically distant populations of *M. pyrifera* might explain the inconsistent evidence for local thermal adaptation in this species. Drifting sporophytes of *M. pyrifera* can remain fertile for up to 125 days [58], permitting high gene flow in areas where ocean currents are strong enough to transport sporophytes over large distances [1]. This holds true for New Zealand: the level of genetic variation within *M. pyrifera* populations is higher than variation between populations, and there is evidence that populations from the regions of Wellington and Stewart Island form a genetic cluster, despite being approximately 900 kms apart, due to gene flow via the Southland Current [77]. High connectivity between kelp populations over large geographic distances could erode adaptation to local-scale selection pressures [9]. Moreover, natural selection depends on genetic mutation and inheritance, which are both inherently random processes. Hence, even when adaptation does occur, it does not result in organisms that are perfectly adapted to their environment.

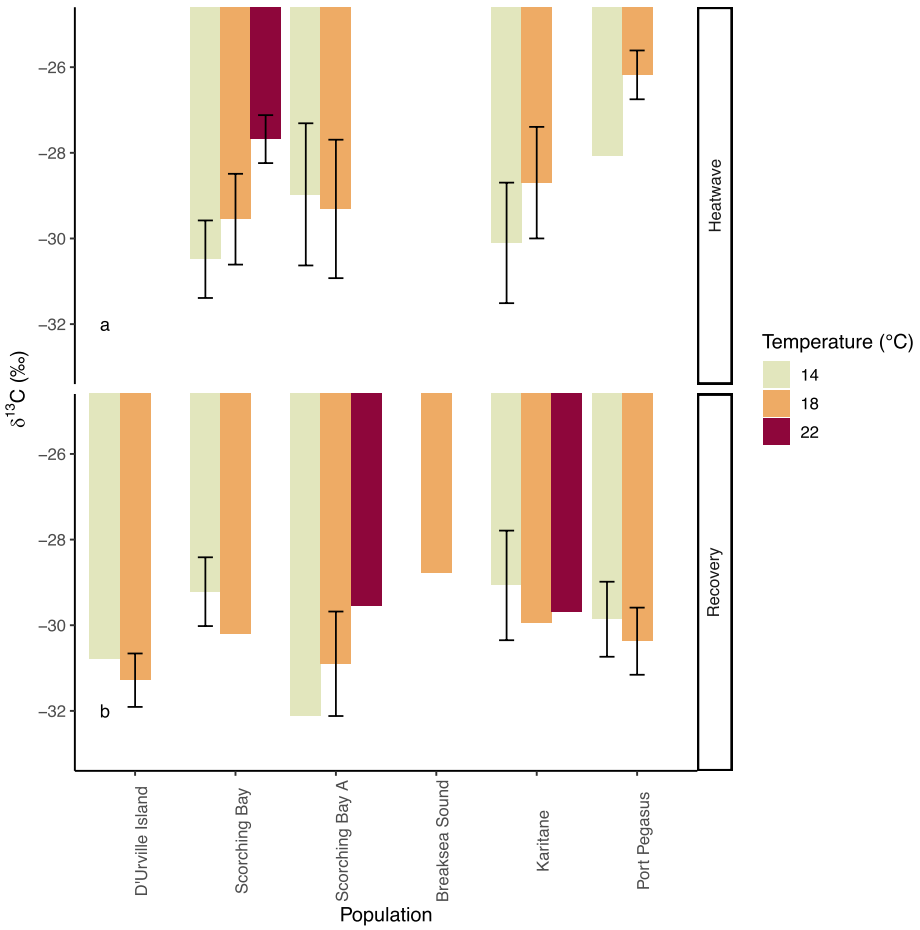


Fig. 9 Mean normalised $\delta^{13}C$ values, with standard error (see Table S5 for *n*), of *Macrocystis pyrifera* blade tissue samples taken: **a** immediately after the heatwave and, **b** at the conclusion of the recovery phase. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

MHWs are likely to have negative impacts on survival and growth of juvenile *M. pyrifera* sporophytes throughout New Zealand. Given the limited evidence for differences in thermal tolerance between populations, the severity of the impacts of MHWs is likely to depend on the characteristics of the MHWs themselves. Strong, rapid-onset MHWs, with large temperature anomalies relative to seasonal averages, could surpass sporophytes' ability to acclimatise, causing widespread reductions in sporophyte health and survival rates, as occurred in the 22 °C heatwave treatment in this experiment. During region-wide MHWs, temperatures within the coastal regions of Marlborough, Wellington, and Fiordland have often exceeded 18 °C and occasionally surpassed 20 °C, whereas temperatures in Otago and Stewart Island have rarely reached 18 °C (Supplementary File 1: Fig. S2). Although the 22 °C treatment used in this experiment is more severe than any region-wide MHW events that have occurred within these regions to date, local-scale data shows that temperature spikes of up to 21.5 °C have occurred at some sites within the Wellington region [72], and mean MHW intensities throughout New Zealand are likely to rise further if greenhouse gas emissions continue to increase [10, 87]. Fiordland has also experienced particularly large temperature anomalies when compared to the other study regions, with several recent events reaching maximum temperatures more than 4 °C above seasonal averages (Supplementary File 1, Figure S4c). These extreme temperatures, coupled with the geographic isolation that limits the flow of new propagules into the region [77], could make *M. pyrifera* populations in Fiordland especially vulnerable to MHWs. Additionally, some populations may experience greater cumulative stress during MHWs if they are affected by other local stressors. For example, *M. pyrifera* populations near highly urbanised areas may be more affected by sedimentation, which limits the penetration of light into coastal habitats [12,

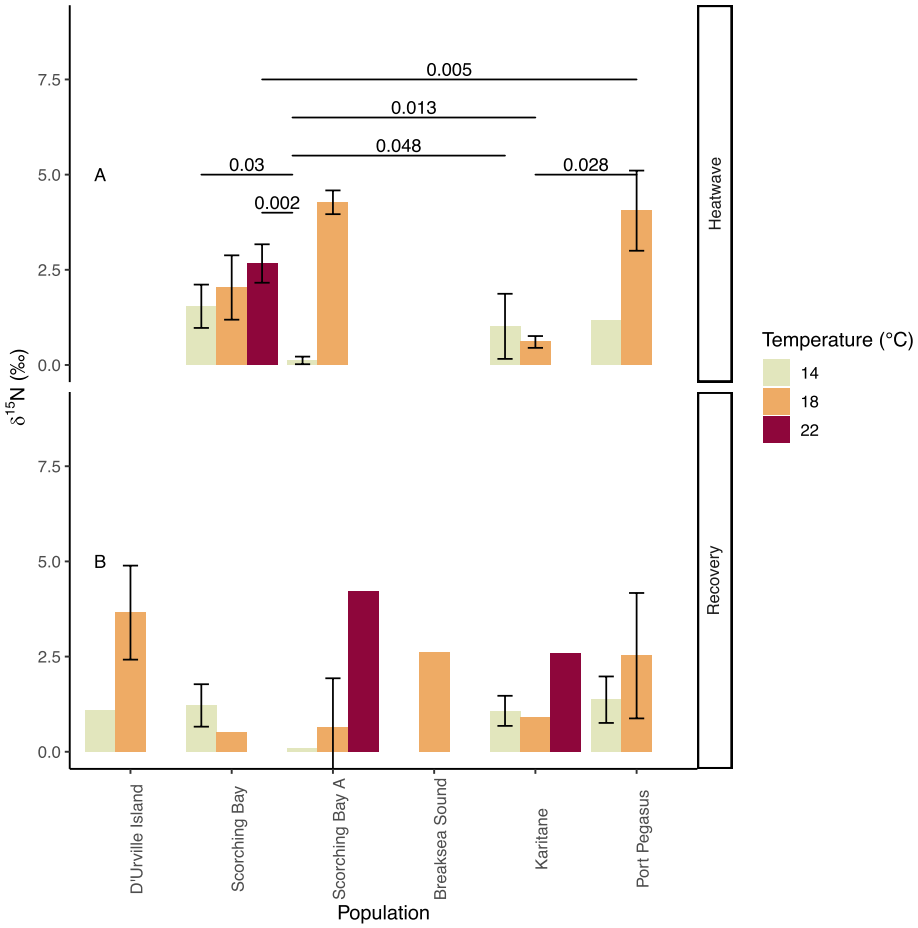


Fig. 10 Mean normalised $\delta^{15}N$ values, with standard error (see Table S5 for *n*), of *Macrocystis pyrifera* blade tissue samples taken: **a** immediately after the heatwave and, **b** at the conclusion of the recovery phase. Populations are arranged along the *x*-axis in order of latitude, from north to south. Scorching Bay A refers to the culture in which gametophytes were stored under white rather than red light

32] and further reduces the resilience of *M. pyrifera* to MHWs [114].

Overall, we found few consistent differences in physiological performance between the Scorching Bay culture in which gametophytes were incubated at 12 °C under red light for 3 years (identified as Scorching Bay in Figs. 2, 3, 4, 5, 6, 7, 8, 9, 10) and the culture in which gametophytes were kept under white light at 15 °C (identified as Scorching Bay A). The Scorching Bay A culture had lower survivorship at 22 °C during the heatwave and slightly higher cumulative survivorship at the end of the experiment, as well as unusually high $\delta^{15}N$ values, but no other parameters were significantly different between the two cultures. Storage of kelp gametophytes under dim, typically red light and cool temperatures is a widely used technique to develop vegetative germ banks for kelp cultivation [51]. This is an effective strategy for preserving gametophytes without the need for cryogenesis; in one study, over 50% of *M. pyrifera* gametophytes remained viable after 5 years in cold storage [7]. Our findings imply that cold storage of kelp gametophytes under red light is unlikely to have significant long-term effects on the viability or resilience of sporophytes produced from these gametophyte cultures, although this conclusion was based on only one source population, so should be considered with some caution. We therefore suggest that cold storage is likely to be an appropriate technique to maintain kelp germ banks for commercial cultivation or preservation of wild genetic diversity.

The impacts of the 22 °C simulated MHW on sporophytes included reductions in survival and linear growth rates, and increases in total carbon content, C:N mass ratio, and $\delta^{15}N$ values, as well as noticeable, but not statistically significant, declines in maximum quantum yield and increases in $\delta^{13}C$ values.

A temperature of 22 °C has previously been identified as a critical threshold for *M. pyrifera* sporophytes from the Wellington region, where survival and growth rates

were greatly reduced after several weeks of exposure to 22 °C [17], whilst higher temperatures caused 100% mortality [16]. Photosynthetic activity also tends to decline at temperatures above 20 °C in *M. pyrifera* sporophytes [17, 42]. The results of this experiment also concur with field data that showed that *M. pyrifera* canopy cover throughout New Zealand remained stable during MHWs at temperatures of up to 18 °C but was greatly reduced at higher temperatures [114]. The increases in sporophyte carbon content, C:N mass ratios, and $\delta^{13}C$ and $\delta^{15}N$ values with increasing temperature found in this study are consistent with previous studies of *M. pyrifera* sporophytes from the Wellington region [17]. Carbon content and C:N mass ratios also tend to increase with temperature in *M. pyrifera* sporophytes from the Americas [15, 43, 82]. It has been suggested that increased temperatures could drive increased uptake and storage of carbon in *M. pyrifera* sporophytes; stored carbon reserves could then be directed towards cellular repair and increased blade growth once temperatures return to optimal values, to compensate for tissue damage and increased erosion during warming events [17]. Transcriptomic analysis could be used to test this hypothesis by analysing whether genes related to carbon acquisition and storage are upregulated at high temperatures.

Survival and recruitment of juvenile kelp is an important stage in population recovery after environmental disturbance [53]. This study demonstrates that MHWs can reduce survival rates of juvenile *M. pyrifera*, potentially delaying or suppressing recovery. Reduced growth rates during MHWs could also leave surviving sporophytes more vulnerable to other stressors. Smaller *M. pyrifera* sporophytes have lower survival rates than larger sporophytes, potentially because they are more susceptible to grazing or less able to compete for light [29]. Thus, small sporophytes could be outcompeted by more heat-tolerant macroalgae, such as fucoids [102], turf algae [44], or other kelp species including *Undaria pinnatifida*, a heat-tolerant invasive kelp which is currently expanding its range in New Zealand [33, 65].

Several kelp restoration projects are currently ongoing or in development throughout New Zealand [45], and these projects could aim to improve the resilience of *M. pyrifera* populations to MHWs. Assisted gene flow, the transfer of genetic material from more diverse or tolerant populations to vulnerable populations, has been proposed as a technique to improve thermal resilience in kelp populations [26, 37]. However, the results of this study do not provide strong evidence for local thermal adaptation among *M. pyrifera* sporophytes in New Zealand, and therefore, assisted gene flow may be of limited efficacy here. Future studies could instead focus on variation in heat tolerance within populations. *M. pyrifera* lineages from a narrow geographic range can vary greatly in thermal tolerance, and measuring this variability to identify and promote heat-resistant genotypes could be an important step towards increasing the resilience of *M. pyrifera* populations [55]. This local-scale approach would also reduce the risks of genetic pollution and maladaptation to other local stressors. In addition, the similar levels of thermal tolerance between genetically distinct populations reared under identical conditions suggest that physiological acclimation and phenotypic plasticity could contribute significantly towards thermal performance in *M. pyrifera* sporophytes. If that is the case, thermal priming could be an effective method of increasing thermal resilience in *M. pyrifera* sporophytes [68], and this would be a worthwhile direction for future study. *M. pyrifera* populations that experience particularly high temperatures or extreme temperature anomalies during MHWs, such as those in the Fiordland region, should be prioritised for this research, as they are likely to be especially vulnerable to MHWs.

5 Conclusions

The findings of this study suggest that *Macrocystis pyrifera* populations throughout New Zealand are likely to be negatively affected by MHWs, especially if average temperature anomalies continue to increase. The similar sensitivity to temperature displayed by all study populations suggests that there is little evidence for local adaptation to thermal stress among *M. pyrifera* sporophytes in New Zealand and assisted gene flow may have limited utility as a strategy to improve thermal tolerance among these populations. Instead, we recommend that future studies focus on understanding variability in thermal tolerance within populations and assessing the potential effectiveness of thermal priming as a technique to promote resilience to MHWs in this ecologically important foundation species.

Abbreviations

MHW	Marine heatwave
PAR	Photosynthetically active radiation
RGR	Relative growth rate
F_v/F_m	Maximum quantum yield of photochemical energy conversion, measured as the ratio of variable fluorescence F_v to maximum fluorescence F_m
$\delta^{13}C$	Ratio of ^{13}C to ^{12}C, expressed in ‰ units
$\delta^{15}N$	Ratio of ^{15}N to ^{14}N, expressed in ‰ units

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s44419-025-00002-z.

Supplementary Material 1. Supplementary tables and figures. Document containing Tables S1–S5 and Figures S1–S5, which contain relevant supporting data that was excluded from the main manuscript to maintain focus on the experimental results. Table and figure legends are provided within the document.

Supplementary Material 2. Raw data. Microsoft Excel workbook containing Tables D1–D4, four spreadsheets containing raw data from this experiment. Table legends are provided within each spreadsheet.

Acknowledgements
The authors wish to thank Isaac Stannard for his help with measurements during the experiment. We also wish to thank Laura Bornemann Santamaría for her help with gametophyte culture and setting up the experiment, and to Graeme Moss and Vonda Cummings for their help with experimental maintenance. We also thank Denisa Berbece, Katie Fenton, Emily Grigg, Ashtyn Isaak, Holly Koch, Maya Korth, Grace Nelles, Alexandra Northmore, Ohad Peleg and Philip Trompke for their feedback and support throughout this process. We also extend our thanks to Ollie Kerr-Hislop and Josette Delgado for their help with stable isotope analysis, and Linda Groenewegen for her help with water nitrogen analysis.

Authors' contributions
ICB, RD, and CEC devised the experiment. DML, MJD and CH were responsible for the collection and storage of kelp gametophytes, including site selection. ICB and YYK carried out the culture of gametophytes and juvenile sporophytes. NB contributed to the setup and maintenance of the experimental system. SJB was responsible for stable isotope analysis. ICB carried out the experiment and performed statistical analysis. CEC, ECK, and RD were responsible for the overall supervision of the project. ICB wrote the manuscript. All authors edited the manuscript.

Funding
This research was supported by funding from the Coastal People, Southern Skies Centre for Research Excellence project to CEC (E4280), a Rutherford Discovery Fellowship to CEC (VUW 1701), the Wellington Community Fund, the Eurofins Foundation, and the Clare Foundation.

Data availability
All data generated or analysed during this study are included in this published article in Supplementary File 2.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
Christopher E. Cornwall is the Editor-in-Chief of Ocean Ecosystems.

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Received: 14 August 2025 Revised: 4 December 2025 Accepted: 8 December 2025
Published online: 02 February 2026

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